Insights Engine Metrics Computation for AI Visibility: A Complete Mathematical Specification

Gravton Labs — Insights Engine

Contents

1 Purpose and Problem Statement

Large Language Model (LLM) answers behave as a *new distribution channel* for user discovery. Unlike classical web search where ranking is visible and link-based, LLM responses contain *implicit rankings* (ordering of entities in text), *implicit prominence* (early vs late mention), *implicit sentiment* toward brands, and sometimes *citations* (or other authority signals).

The goal of this Insights Engine is to compute robust, interpretable, and comparable metrics that quantify a brand's **AI Visibility** relative to competitors across:

- a prompt library representing real user intents,
- multiple LLM platforms/models,
- repeated generations per prompt (to reduce sampling noise),
- time windows for trending and smoothing.

We define a core **Visibility score** along with supporting diagnostics: **Presence Rank**, **Share of Voice (SoV)**, **Position metrics**, **Sentiment metrics**, and **Citation Share**. We also specify an optional **cross-channel composite visibility** that extends beyond LLM answers into broader sources (e.g., search, social, news) using prominence and recency decay.

2 Core Data Model and Notation

2.1 Sets, Indices, and Time Window

Let:

$$P = \{p_1, \dots, p_{|P|}\} \quad \text{(prompt library; each prompt approximates a user intent)}, \tag{1}$$
$$M = \{m_1, \dots, m_{|M|}\} \quad \text{(LLM platforms/models)}, \tag{2}$$
$$B = \{b_1, \dots, b_{|B|}\} \quad \text{(brands/entities; includes the focal brand + competitors)}, \tag{3}$$
$$G = \{1, \dots, K\} \quad \text{(generation index; repeated runs per prompt-model pair)}, \tag{4}$$
$$\mathcal{T} = [t_{\min}, t_{\max}] \quad \text{(analysis time window; e.g., last 7/30 days)}. \tag{5}$$

Why we need K generations. LLM outputs are stochastic (temperature, sampling, retrieval variance). Running $K \geq 1$ generations for each (p, m) allows estimating expected metrics and reducing variance.

2.2 Weights [TBD]

We introduce two weight families:

Prompt weight w_p.

$$w_p \geq 0 \quad \forall p \in P. \tag{6}$$

Interpretation: w_p captures the real-world importance of prompt p (e.g., estimated user prompt volume, business priority, or market size). If prompt volumes are unknown, set $w_p = 1$ initially.

Model weight α_m.

$$\alpha_m \geq 0 \quad \forall m \in M, \qquad \sum_{m \in M} \alpha_m = 1. \tag{7}$$

Interpretation: α_m captures exposure probability or strategic importance of model m. Two common constructions:

- **Market-share weighting:** α_m proportional to estimated global usage share.
- **Client-traffic weighting:** α_m proportional to observed AI-referral traffic from model m.

2.3 Per-Generation Extracted Variables

For each prompt $p \in P$, model $m \in M$, generation $g \in G$ with response text $R_{p,m,g}$ generated within time window $\mathcal{T}$, we compute per brand $b \in B$:

Binary Presence Indicator.

$$\mathrm{Pres}_{b,p,m,g} \in \{0, 1\}, \tag{8}$$

where $\mathrm{Pres}_{b,p,m,g} = 1$ iff brand b is mentioned at least once in $R_{p,m,g}$ after entity resolution (alias mapping + disambiguation).

Mention Frequency.

$$f_{b,p,m,g} \in \mathbb{Z}_{\geq 0} \tag{9}$$

equals the count of mentions of b in $R_{p,m,g}$.

First Mention Position (Token Index). Let $T_{p,m,g}$ be the total token count of the response (using a chosen tokenizer):

$$T_{p,m,g} \in \mathbb{Z}_{>0}. \tag{10}$$

Define:

$$\mathrm{pos}_{b,p,m,g} \in \{1, \ldots, T_{p,m,g}\} \quad \text{if } \mathrm{Pres}_{b,p,m,g} = 1, \tag{11}$$

as the index of the first token where the canonical mention of b occurs. If absent, we leave pos undefined (or set to $+\infty$ purely for engineering convenience).

Entity Rank (Order Among Brands). Define $\mathrm{rank}_{b,p,m,g} \in \{1, 2, \ldots\}$ when present. A robust definition is:

$$\mathrm{rank}_{b,p,m,g} := 1 + \#\{b' \in B \,:\, \mathrm{Pres}_{b',p,m,g} = 1 \wedge \mathrm{pos}_{b',p,m,g} < \mathrm{pos}_{b,p,m,g}\}. \tag{12}$$

Thus rank is determined by *earliest appearance*. This is stable across formats (lists, paragraphs) and aligns with the intuitive idea of prominence.

Aspect-based Sentiment. Let $S_{b,p,m,g} \in [-1, 1]$ be sentiment toward brand b in context of prompt p and response $R_{p,m,g}$. We emphasize **aspect-based** sentiment: sentiment toward the brand, not global tone.

A typical operational construction:

1. Extract a brand-centered context window $\mathrm{Ctx}_{b,p,m,g}$ (e.g., sentence containing mention ± 1 sentence, or ± 50 tokens).
2. Apply a classifier producing probabilities $P(\mathrm{pos}), P(\mathrm{neu}), P(\mathrm{neg})$.
3. Convert to scalar: $S = P(\mathrm{pos}) - P(\mathrm{neg})$.
4. If classifier uncertainty is high, optionally use an LLM fallback prompt that outputs a scalar in $[-1, 1]$.

Citations / Authority Signals. Let $C_{b,p,m,g} \in \mathbb{Z}_{\geq 0}$ be the number of citations in $R_{p,m,g}$ attributed to brand b. Attribution requires a rule, e.g.:

- citation appears within a context window near a brand mention, or
- citation domain is owned/controlled by brand (brand domain mapping), or
- explicit structured citation-to-claim mapping when available.

2.4 Important Engineering Clarifications (Mathematically Relevant)

- **Entity resolution is mandatory:** map aliases/synonyms to a canonical brand id. Otherwise presence/frequency/rank are wrong.
- **Rank definition must be consistent:** use earliest occurrence (token position) by default.
- **Conditioning on presence:** for averages of sentiment/position/rank, exclude absent cases; otherwise missing mentions bias metrics.
- **Response length variation:** token-based position measures should be normalized by $T_{p,m,g}$.

2.5 Weighted Sum Denominators

We frequently need normalization constants:

$$W_P := \sum_{p \in P} w_p, \tag{13}$$

$$W_P^{(b,m)} := \sum_{p \in P} w_p \cdot \mathbf{1}\{\exists g : \mathrm{Pres}_{b,p,m,g} = 1\} \quad \text{(prompt-weight mass where brand appears at least once for model} \tag{14}$$

If you compute metrics per-generation, replace the indicator with $\mathrm{Pres}_{b,p,m,g}$ accordingly.

3 Per-Generation Mention Strength Components

The engine's core idea is that a brand's strength in an answer depends on:

- **Whether it appears** (presence),
- **How early it appears** (rank or position),
- **How often it appears** (frequency).

3.1 Rank Weight

Define the rank weight:

$$\mathrm{RankW}_{b,p,m,g} := \begin{cases} \frac{1}{\mathrm{rank}_{b,p,m,g}}, & \mathrm{Pres}_{b,p,m,g} = 1, \\ 0, & \mathrm{Pres}_{b,p,m,g} = 0. \end{cases} \tag{15}$$

Properties:

- RankW $\in (0, 1]$ when present.
- Strongly rewards rank 1; diminishing for lower ranks.
- Interpretability: "You are #1" directly increases score.

3.2 Frequency Weight

Define the frequency weight using logarithmic compression:

$$\mathrm{FreqW}_{b,p,m,g} := \log(1 + f_{b,p,m,g}). \tag{16}$$

Properties:

- FreqW $= 0$ when $f = 0$.
- Additional mentions contribute diminishing increments.

3.3 Mention Score (Per Generation)

Define mention score as:

$$\mathrm{MS}_{b,p,m,g} := \mathrm{Pres}_{b,p,m,g} \cdot \mathrm{RankW}_{b,p,m,g} \cdot \mathrm{FreqW}_{b,p,m,g}. \tag{17}$$

Interpretation:

- If absent, MS $= 0$.
- If present and early, rank term is large.
- If present and repeated, frequency term increases moderately.

3.4 Alternative: Token-Normalized Position Score

Sometimes rank is too coarse; define a continuous position score:

$$\mathrm{PosScore}_{b,p,m,g} := \begin{cases} 1 - \frac{\mathrm{pos}_{b,p,m,g}}{T_{p,m,g}}, & \mathrm{Pres}_{b,p,m,g} = 1, \\ 0, & \mathrm{Pres}_{b,p,m,g} = 0. \end{cases} \tag{18}$$

Properties:

- PosScore $\in [0, 1)$.
- Captures early mention in long answers more finely than rank.
- Sensitive to response length and tokenization; use for diagnostics.

3.5 Choice Guidance: Rank vs Position

- Use **RankW** for competitive scoring and stakeholder-friendly reporting.
- Use **PosScore** for debugging and deeper analysis (why rank changed, why mention moved).

4 Prompt-Level Aggregation Across Generations

Because we run K generations per (p, m), we first aggregate per prompt and model to reduce stochastic noise.

4.1 Prompt-Model Expected Mention Score

Define expected mention score:

$$\overline{\mathrm{MS}}_{b,p,m} := \frac{1}{K}\sum_{g=1}^{K}\mathrm{MS}_{b,p,m,g}. \tag{19}$$

Similarly, define expected rank, frequency, position, sentiment, citations:

$$\overline{f}_{b,p,m} := \frac{1}{K}\sum_{g=1}^{K} f_{b,p,m,g}, \tag{20}$$

$$\overline{C}_{b,p,m} := \frac{1}{K}\sum_{g=1}^{K} C_{b,p,m,g}. \tag{21}$$

Conditioned averages (recommended). For rank, position, and sentiment, average *only when present*:

$$\overline{\mathrm{rank}}_{b,p,m} := \frac{\sum_{g=1}^{K}\mathrm{rank}_{b,p,m,g}\cdot \mathrm{Pres}_{b,p,m,g}}{\sum_{g=1}^{K}\mathrm{Pres}_{b,p,m,g}+\epsilon}, \tag{22}$$

$$\overline{\mathrm{PosScore}}_{b,p,m} := \frac{\sum_{g=1}^{K}\mathrm{PosScore}_{b,p,m,g}}{K}, \tag{23}$$

$$\overline{S}_{b,p,m} := \frac{\sum_{g=1}^{K} S_{b,p,m,g}\cdot \mathrm{Pres}_{b,p,m,g}}{\sum_{g=1}^{K}\mathrm{Pres}_{b,p,m,g}+\epsilon}, \tag{24}$$

where $\epsilon > 0$ is a tiny constant to avoid division by zero in purely absent cases (engineering safeguard). If absent in all generations, treat the conditioned statistic as undefined for reporting (or set to 0 with a missing flag).

5 Visibility Metric (Core Presence/Visibility Score)

5.1 Prompt-Level Visibility Contribution

Define the prompt-level contribution for brand b under prompt p and model m:

$$\mathrm{Vis}_{b,p,m} := w_p \cdot \overline{\mathrm{MS}}_{b,p,m}. \tag{25}$$

Interpretation: higher-value prompts (high w_p) contribute more to overall visibility.

5.2 Model-Level Visibility

Aggregate prompt contributions within a model:

$$\mathrm{Vis}_{b,m} := \sum_{p\in P}\mathrm{Vis}_{b,p,m} \;=\; \sum_{p\in P} w_p \cdot \overline{\mathrm{MS}}_{b,p,m}. \tag{26}$$

Optional scale invariance (weighted average). If you want model-level visibility not to scale with the size of prompt library, normalize:

$$\mathrm{Vis}^{\mathrm{avg}}_{b,m} := \frac{\sum_{p\in P} w_p\cdot \overline{\mathrm{MS}}_{b,p,m}}{\sum_{p\in P} w_p}. \tag{27}$$

In dashboards, either choice is valid; the normalized version is easier to compare across varying prompt sets.

5.3 Cross-Model Visibility (Raw)

Combine models using exposure weights:

$$\mathrm{Vis}_b^{\mathrm{raw}} := \sum_{m \in M} \alpha_m \cdot \mathrm{Vis}_{b,m}. \tag{28}$$

5.4 Competitive Normalization to 0–100

To produce a relative visibility score on a 0–100 scale:

$$\mathrm{Visibility}_b := 100 \cdot \frac{\mathrm{Vis}_b^{\mathrm{raw}}}{\max_{k \in B} \mathrm{Vis}_k^{\mathrm{raw}} + \epsilon}. \tag{29}$$

Properties:

- The top brand in set B gets approximately 100.
- Others scale proportionally.
- ϵ prevents division-by-zero if all raw visibilities are 0 (rare; occurs if no brands are ever mentioned).

5.5 Temporal Smoothing (Exponential Smoothing)

For stability over time (e.g., daily computations), define:

$$\mathrm{Visibility}_{b,t}^{\mathrm{final}} := \lambda \cdot \mathrm{Visibility}_{b,t} + (1 - \lambda) \cdot \mathrm{Visibility}_{b,t-1}^{\mathrm{final}}, \tag{30}$$

where $\lambda \in (0, 1)$ is a smoothing constant. A commonly used setting:

$$\lambda = 0.7 \quad \Rightarrow \quad \text{“70\% current, 30\% previous”}. \tag{31}$$

5.6 Presence Rank (Dense Ranking)

Compute ranks based on $\mathrm{Visibility}_{b,t}^{\mathrm{final}}$ descending. Define dense rank:

$$\mathrm{Rank}_{b,t}^{\mathrm{presence}} := 1 + \#\{k \in B : \mathrm{Visibility}_{k,t}^{\mathrm{final}} > \mathrm{Visibility}_{b,t}^{\mathrm{final}}\}. \tag{32}$$

Dense ranking ensures ties share rank without gaps.

6 Share of Voice (SoV)

SoV represents a brand's share of total visibility mass within the competitor set.

6.1 SoV Definition

Define total raw visibility mass:

$$\mathrm{TotVis}^{\mathrm{raw}} := \sum_{k \in B} \mathrm{Vis}_k^{\mathrm{raw}}. \tag{33}$$

Then:

$$\mathrm{SoV}_b := 100 \cdot \frac{\mathrm{Vis}_b^{\mathrm{raw}}}{\mathrm{TotVis}^{\mathrm{raw}} + \epsilon}. \tag{34}$$

Interpretation: among all competitor visibility, what fraction belongs to brand b.

6.2 SoV Rank (Dense)

$$\mathrm{Rank}_b^{\mathrm{SoV}} := 1 + \#\{k \in B : \mathrm{SoV}_k > \mathrm{SoV}_b\}. \tag{35}$$

7 Position Metrics (Prominence in Answers)

Position metrics answer: when a brand appears, how early does it appear?

7.1 Average Rank Per Model (Conditioned on Presence)

Define, for each model m:

$$\mathrm{AvgRank}_{b,m} := \frac{\sum_{p\in P} w_p \cdot \overline{\mathrm{rank}}_{b,p,m} \cdot \mathbf{1}\{\text{brand appears for } (b,p,m)\}}{\sum_{p\in P} w_p \cdot \mathbf{1}\{\text{brand appears for } (b,p,m)\} + \epsilon}. \tag{36}$$

A practical indicator:

$$\mathbf{1}\{\text{brand appears for } (b,p,m)\} := \mathbf{1}\left\{\sum_{g=1}^{K} \mathrm{Pres}_{b,p,m,g} > 0\right\}.$$

Interpretation: weighted average of rank, only among prompts where the brand appears at least once.

7.2 Cross-Model Average Rank

Aggregate across models:

$$\mathrm{AvgRank}_b := \sum_{m\in M} \alpha_m \cdot \mathrm{AvgRank}_{b,m}. \tag{37}$$

7.3 Position Rank (Ascending Dense)

Lower average rank is better (closer to #1):

$$\mathrm{Rank}_b^{\mathrm{position}} := 1 + \#\{k \in B : \mathrm{AvgRank}_k < \mathrm{AvgRank}_b\}. \tag{38}$$

7.4 Optional: Average Token-Normalized Position

If using PosScore:

$$\mathrm{AvgPosScore}_{b,m} := \frac{\sum_{p\in P} w_p \cdot \overline{\mathrm{PosScore}}_{b,p,m}}{\sum_{p\in P} w_p}, \qquad \mathrm{AvgPosScore}_b := \sum_{m\in M} \alpha_m \cdot \mathrm{AvgPosScore}_{b,m}. \tag{39}$$

8 Sentiment Metrics

Sentiment metrics answer: when the brand is mentioned, is it described positively, neutrally, or negatively?

8.1 Overall Weighted Sentiment Score

Define a presence-conditioned, prompt-weighted sentiment per model:

$$\mathrm{Sent}_{b,m} := \frac{\sum_{p\in P} w_p \cdot \overline{S}_{b,p,m} \cdot \mathbf{1}\{\sum_g \mathrm{Pres}_{b,p,m,g} > 0\}}{\sum_{p\in P} w_p \cdot \mathbf{1}\{\sum_g \mathrm{Pres}_{b,p,m,g} > 0\} + \epsilon}. \tag{40}$$

Cross-model aggregation:

$$\mathrm{Sent}_b := \sum_{m\in M} \alpha_m \cdot \mathrm{Sent}_{b,m}. \tag{41}$$

Range: $\mathrm{Sent}_b \in [-1, 1]$.

8.2 Sentiment Buckets: Positive / Neutral / Negative

Define indicator functions for bucket membership at prompt-model level using the aggregated sentiment:

$$\mathbf{1}^{+}_{b,p,m} := \mathbf{1}\{\overline{S}_{b,p,m} > 0\} \cdot \mathbf{1}\{\sum_g \mathrm{Pres}_{b,p,m,g} > 0\}, \tag{42}$$

$$\mathbf{1}^{0}_{b,p,m} := \mathbf{1}\{\overline{S}_{b,p,m} = 0\} \cdot \mathbf{1}\{\sum_g \mathrm{Pres}_{b,p,m,g} > 0\}, \tag{43}$$

$$\mathbf{1}^{-}_{b,p,m} := \mathbf{1}\{\overline{S}_{b,p,m} < 0\} \cdot \mathbf{1}\{\sum_g \mathrm{Pres}_{b,p,m,g} > 0\}. \tag{44}$$

Then compute weighted percentages per model:

$$\mathrm{PosPct}_{b,m} := \frac{\sum_{p\in P} w_p \cdot \mathbf{1}^{+}_{b,p,m}}{\sum_{p\in P} w_p \cdot \mathbf{1}\{\sum_g \mathrm{Pres}_{b,p,m,g} > 0\} + \epsilon}, \tag{45}$$

$$\mathrm{NegPct}_{b,m} := \frac{\sum_{p\in P} w_p \cdot \mathbf{1}^{-}_{b,p,m}}{\sum_{p\in P} w_p \cdot \mathbf{1}\{\sum_g \mathrm{Pres}_{b,p,m,g} > 0\} + \epsilon}, \tag{46}$$

$$\mathrm{NeuPct}_{b,m} := 1 - \mathrm{PosPct}_{b,m} - \mathrm{NegPct}_{b,m}. \tag{47}$$

Aggregate across models:

$$\mathrm{PosPct}_b := \sum_{m\in M} \alpha_m \cdot \mathrm{PosPct}_{b,m}, \tag{48}$$

$$\mathrm{NegPct}_b := \sum_{m\in M} \alpha_m \cdot \mathrm{NegPct}_{b,m}, \tag{49}$$

$$\mathrm{NeuPct}_b := 1 - \mathrm{PosPct}_b - \mathrm{NegPct}_b. \tag{50}$$

8.3 Sentiment Opportunities (Actionable Themes)

For each mention context $\mathrm{Ctx}_{b,p,m,g}$, extract a topic/phrase label τ (e.g., "pricing", "delivery speed", "support quality") via clustering or classification. Let $\mathcal{U}_b$ be the set of sentiment themes for brand b.

Define weighted theme count:

$$\mathrm{ThemeCount}_{b,\tau} := \sum_{m\in M}\sum_{p\in P} \alpha_m w_p \cdot \mathbf{1}\{\tau \text{ extracted from } \mathrm{Ctx}_{b,p,m}\} \cdot \mathbf{1}\{\sum_g \mathrm{Pres}_{b,p,m,g} > 0\}. \tag{51}$$

You may also compute per-theme sentiment:

$$\mathrm{ThemeSent}_{b,\tau} := \frac{\sum_{m,p} \alpha_m w_p \cdot \overline{S}_{b,p,m} \cdot \mathbf{1}\{\tau\}}{\sum_{m,p} \alpha_m w_p \cdot \mathbf{1}\{\tau\} + \epsilon}. \tag{52}$$

Then:

- Top opportunities: themes with large ThemeCount and low ThemeSent.
- Strengths: themes with large ThemeCount and high ThemeSent.

9 Citation Share Metrics

Citations measure authority and support in model responses.

9.1 Citation Mass (Raw)

Compute per model:

$$\mathrm{Cit}_{b,m} := \sum_{p \in P} w_p \cdot \overline{C}_{b,p,m}. \tag{53}$$

Cross-model:

$$\mathrm{Cit}_b^{\mathrm{raw}} := \sum_{m \in M} \alpha_m \cdot \mathrm{Cit}_{b,m}. \tag{54}$$

9.2 Citation Share

$$\mathrm{CitShare}_b := 100 \cdot \frac{\mathrm{Cit}_b^{\mathrm{raw}}}{\sum_{k \in B} \mathrm{Cit}_k^{\mathrm{raw}} + \epsilon}. \tag{55}$$

9.3 Citation Rank (Dense)

$$\mathrm{Rank}_b^{\mathrm{cit}} := 1 + \#\{k \in B : \mathrm{CitShare}_k > \mathrm{CitShare}_b\}. \tag{56}$$

9.4 Citation Availability Rate (Recommended Diagnostic)

Some models frequently omit citations. Track:

$$\mathrm{CitAvail}_m := \frac{\sum_{p,g} \mathbf{1}\{\text{response } R_{p,m,g} \text{ contains any citations}\}}{|P| \cdot K}. \tag{57}$$

This prevents misinterpretation when citation volume changes due to model behavior rather than brand performance.

10 Model Weight Construction α_m

10.1 Market Share Weighting

If you have external usage estimates $u_m \geq 0$, define:

$$\alpha_m := \frac{u_m}{\sum_{j \in M} u_j}. \tag{58}$$

10.2 Client-Specific Traffic Weighting

If you have client observed referral traffic $r_m \geq 0$:

$$\alpha_m := \frac{r_m}{\sum_{j \in M} r_j}. \tag{59}$$

10.3 Hybrid Weighting (Optional)

Blend market and client weights:

$$\alpha_m := \beta \cdot \alpha_m^{\mathrm{market}} + (1 - \beta) \cdot \alpha_m^{\mathrm{client}}, \quad \beta \in [0, 1]. \tag{60}$$

11 Cross-Channel Composite Visibility (Optional Extension)

Beyond LLM answers, the doc proposes a more general visibility computed from *mentions* in many sources: search, social, news, reviews, etc. We formalize that here.

11.1 Mention Events

Let $\mathcal{E}$ be the set of mention events. Each event $e \in \mathcal{E}$ is a tuple:

$$e = (\text{source}, \text{document}, \text{timestamp}, b, \text{placement}, \text{reach}, \text{credibility}, \dots)$$

within time window $\mathcal{T}$.

11.2 Prominence / Placement Weight

Let $\pi(e) \in [0, 1]$ be a placement prominence weight. A default rubric:

$$\pi(e) = \begin{cases} 1.0, & \text{Title mention,} \\ 0.7, & \text{H1 mention,} \\ 0.6, & \text{H2 mention,} \\ 0.5, & \text{H3 mention,} \\ 0.5, & \text{Body mention,} \\ 0.3, & \text{Footer/low-salience mention.} \end{cases} \tag{61}$$

11.3 Recency Decay

Let $\Delta(e)$ be days since event timestamp:

$$\Delta(e) = \text{days}(t_{\text{max}} - t(e)).$$

Define exponential half-life decay:

$$d(e) := 2^{-\Delta(e)/h}, \tag{62}$$

where $h > 0$ is the half-life in days (default $h = 7$ days). Interpretation: every h days, event contribution halves.

11.4 Reach and Credibility

Let:

$$r(e) \geq 0 \quad \text{(reach proxy; e.g., estimated impressions, follower count, traffic)}, c(e) \in [0, 1] \quad \text{(credibility score; e.g} \tag{63}$$

11.5 Raw Composite Visibility

Define raw cross-channel visibility:

$$V_b^{\text{raw}} := \sum_{e \in \mathcal{E}:\text{brand}(e)=b} r(e) \cdot c(e) \cdot \pi(e) \cdot d(e). \tag{64}$$

11.6 Normalization

Two common normalizations:

1. **Competitive max normalization:**

$$V_b := \frac{V_b^{\text{raw}}}{\max_{k \in B} V_k^{\text{raw}} + \epsilon}. \tag{65}$$

2. **Total-mass normalization (share):**

$$V_b^{\text{share}} := \frac{V_b^{\text{raw}}}{\sum_{k \in B} V_k^{\text{raw}} + \epsilon}. \tag{66}$$

12 Implementation-Critical Edge Cases and Definitions

12.1 Entity Resolution and Canonical Mapping

Define an alias dictionary $\mathcal{A}_b$ for each brand b. A mention string s maps to brand b if:

$$s \in \mathcal{A}_b$$

or via NER + fuzzy matching + disambiguation. Without this, $\text{Pres}, f, \text{pos}, \text{rank}$ are invalid.

12.2 Handling Ties in Rank

If two brands appear at the same token position (rare; usually impossible), define a tie-break:

- alphabetical by canonical id, or
- earliest character offset.

Ensure deterministic ranking.

12.3 Presence Conditioning: Why It Matters

For metrics like average rank and sentiment, including absent prompts as zeros can create artifacts:

- It confounds "not mentioned" with "neutral" sentiment.
- It penalizes brands for prompt coverage in a way that double-counts presence effects (already captured in visibility).

Thus we condition sentiment and rank on presence.

12.4 Tokenization Choice

pos and $T_{p,m,g}$ depend on tokenizer. Use a consistent tokenizer per model or a unified tokenizer for comparability. If unified, treat position as an approximate proxy across models.

12.5 Smoothing Window Alignment

If metrics are computed daily with a trailing window $\mathcal{T}$:

- "current" refers to window ending at day t,
- "previous" refers to window ending at day $t - 1$ (or $t - \delta$).

Be explicit in reporting.

13 Summary of Outputs (What the Engine Produces)

For each brand b in competitor set B, the Insights Engine outputs at time t:

- $\text{Visibility}_{b,t}^{\text{final}}$: normalized 0–100 visibility score (smoothed)
- $\text{Rank}_{b,t}^{\text{presence}}$: dense rank by visibility
- $\text{SoV}_{b,t}$ and $\text{Rank}_{b,t}^{\text{SoV}}$: share of voice and rank
- $\text{AvgRank}_{b,t}$ and $\text{Rank}_{b,t}^{\text{position}}$: prominence rank (lower is better)
- $\text{Sent}_{b,t}$: sentiment scalar in $[-1, 1]$
- $\text{PosPct}_{b,t}, \text{NeuPct}_{b,t}, \text{NegPct}_{b,t}$: sentiment distribution
- $\text{CitShare}_{b,t}$ and $\text{Rank}_{b,t}^{\text{cit}}$: citation share and rank
- Sentiment theme tables: $\text{ThemeCount}_{b,\tau}$, $\text{ThemeSent}_{b,\tau}$ for opportunities
- Model diagnostics: CitAvail_m (and optionally response length stats)

14 Recommended Defaults (Operational Settings)

- Generations per prompt-model: $K \in \{3, 5\}$ (tradeoff cost vs variance)
- Prompt weights: start $w_p = 1$; later use estimated market prompt volume
- Model weights: start with market-share estimates; customize by client traffic
- Visibility: use RankW $\times$ log-frequency weighting
- Smoothing: $\lambda = 0.7$ daily (or tune per volatility)
- Recency half-life (cross-channel): $h = 7$ days (tune by vertical)

15 Conclusion

This specification defines a complete mathematical framework for AI visibility scoring across prompts and models. The framework intentionally separates:

- **Visibility mass** (presence + rank + frequency),
- **Relative competitiveness** (normalization, SoV),
- **Prominence diagnostics** (average rank / position),
- **Tone/brand health** (sentiment),
- **Authority/credibility** (citations),
- **Cross-channel expansion** (prominence + reach + recency decay).

This yields an interpretable, actionable insights engine suitable for dashboarding, alerting, and optimization loops.